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FULL PAPER

Influence of substrate temperature on the structural and optoelectronic properties of ZnMgO:Al coatings deposited by radio frequency magnetron sputtering

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ZnMgO:Al coatings were prepared by radio frequency (RF) magnetron sputtering on glass substrates, and the effects of the substrate temperature on the structural, electrical, and optical properties of ZnMgO:Al coatings were studied. The ZnMgO:Al coatings had a hexagonal wurtzite crystal structure oriented along the *c*-axis, regardless of the substrate temperature. Increasing the substrate temperature increased the grain sizes of the coatings and thus their Hall mobilities. The high substrate temperature also promoted the doping concentration efficiency of Al^{3+} ions, leading to enhanced carrier concentration. The optimal deposition was obtained at 400 °C, which led to the lowest resistivity ($2.82 \times 10^{-3}\ \Omega$ cm) and 91 % transmission in the visible range. The optical bandgap increased to 3.632 eV as the substrate temperature increased to 400 °C. Wide bandgap, highly transparent, and conductive ZnMgO:Al coatings can be used as electrodes for ultraviolet photovoltaic applications.
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1. Introduction

Transparent conducting oxides (TCOs) such as Sn-doped In_2O_3 (ITO), SnO_2, and ZnO have attracted interest due to their applications in display devices, solar cells, and optoelectronic devices.[1)] Among them, ITO is the most commonly used TCOs due to its low resistivity and high transmittance in the visible region. In addition, it offers low cost, nontoxicity, and stability in the hydrogen atmosphere. Similarly, ZnO has also attracted attention as a powerful candidate for transparent ITO electrodes. ZnO films doped with Al, In, Ga, Cu, or Cd have been studied. In particular, Al doping can induce pronounced n-type conductivity because Al^{3+} ions readily act as electron donors in ZnO due to the smaller ionic radius of highly reactive Al.[2)–4)]

Several techniques have been employed to prepare Al-doped ZnO (AZO) thin films, such as pulsed laser deposition, magnetron sputtering, chemical vapor deposition, and sol–gel processing. Magnetron sputtering is an industrially powerful deposition method for producing TCO coatings as it offers large-area coating at high deposition rates and can be used to coat a wide range of materials. Two primary sputtering methods can be used for the Al doping of ZnO: co-sputtering using ZnO and Al targets and conventional sputtering using the $ZnO + Al_2O_3$ target. The latter is more suitable for industrial processes due to the simplicity of the growth process. Because the structures and properties of coatings depend on processing conditions, the substrate temperature plays a crucial role in the growth of AZO films. It has been shown to improve the crystallinity and electrical properties of AZO films.[5)–7)]

TCOs with high optical transparency in the ultraviolet (UV) range have been used in solar cells,[8)–10)] photodetectors,[11),12)] and luminescent applications.[13)] In addition, wide-bandgap TCO heterostructures have been used to enhance mobility by modulation.[14)] In these structures, conduction band electrons in wider-bandgap layers transfer across heterojunction interfaces to intrinsic ZnO layers. The bandgap of ZnO can be widened by Mg doping. However, Mg^{2+} substitution cannot generate any free carriers due to the same valence states of Mg^{2+} and Zn^{2+}. To produce a high-conducting ZnO with a large bandgap and enhance its device performance, Al and Mg have been incorporated to enhance its carrier concentration and expand its bandgap, respectively.

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Table 1. Comparison of optoelectronic performance for donor-doped ZnMgO films obtained at various sputtering parameters

Target	Mg-doping concentration (at.%)	Substrate temperature (°C)	Annealing temperature (°C)	Visible light transmittance (%)	Bandgap (eV)	Electrical resistivity (Ω cm)	Reference
ZnMg (10 at.% Mg) and Al	N/A	300	No	N/A	~3.5	2.06×10^{-1}	15)
ZnMgGaO (15 at.% Mg/2 at.% Ga)	N/A	350	No	~90	~3.6	$\sim 4.0 \times 10^{-2}$	16)
ZnMgO (21 at.% MgO), ZnF_2, and MgF_2	N/A	RT	650 (30 min in a H_2/Ar)	~86 % (380–780 nm)	3.79	1.26×10^{-1}	17)
ZnAlO (2 wt.% Al_2O_3) and Mg	6.1	350	No	~75 %	~3.9	$\sim 3.4 \times 10^{-2}$	18)
ZnMgAlO (9 wt.% MgO/2 wt.% Al_2O_3)	8.19	400	No	~91 (380–780 nm)	3.632	2.82×10^{-3}	Present work

RT: Room temperature.

So far, few reports have addressed using sputtering technology to prepare donor-doped ZnMgO coatings for enhanced resistivity and optical bandgap widening (**Table 1**). The optimal optoelectronic properties for donor-doped ZnMgO coatings were achieved using a high growth temperature or post-annealing temperature. In these reports, ZnMgO thin films were prepared by reactive magnetron co-sputtering technology using multiple targets, which cause poor stability and composition uniformity. Few reports dealt with the deposition of donor-doped ZnMgO films using a single ceramic target.

In this study, ZnMgO:Al coatings were prepared by radio frequency (RF) magnetron sputtering on glass substrates at different substrate temperatures using a single ceramic target. The $Zn_{0.83}Mg_{0.17}O$ coating has excellent application potential in solar cells[9),10)] and exhibits luminescence[13)] and high-mobility two-dimensional electron gas.[14)] To achieve the required chemical composition, we selected ZnO containing 9.2 wt.% MgO. To optimize electrical performance, 2 wt.% Al was incorporated into $Zn_{0.83}Mg_{0.17}O$, forming a sputtering target involving 89 wt.% ZnO, 9 wt.% MgO, and 2 wt.% Al_2O_3. The structure and optoelectronic properties of the ZnMgO:Al coatings were investigated by varying the substrate temperature.

2. Experimental procedure

ZnMgO:Al coatings were fabricated in a high-vacuum dual-target sputter deposition system on 1 cm × 1 cm glass substrates using ceramic targets (diameter: 3 in). The ceramic targets were prepared by conventional solid-state reaction. A mixture of high-purity ZnO (99.99 %), Al_2O_3 (99.99 %), and MgO (99.99 %) powders was pressed using cold isostatic pressing and sintered at 950 °C for 6 h in a furnace. The ceramic target with 9 wt.% MgO and 2 wt.% Al_2O_3 was employed, and the ceramic target with 9.2 wt.% MgO was used for reference. Prior to deposition, the substrates were ultrasonically cleaned in an acetone and methanol bath for 10 min to improve the adhesion of the film. The substrates were subsequently blasted with dry nitrogen before placing in the sputtering chamber. The deposition chamber was initially evacuated down to a pressure of less than 4.0×10^{-4} Pa, and the chamber pressure was maintained at 4.0×10^{-1} Pa during sputtering by flowing Ar gas at a constant rate of 40 sccm. During deposition, the RF power was fixed at 125 W, and the substrate temperature was varied in the range of room temperature (RT)–400 °C. At temperatures higher than 500 °C, the glass substrate was prone to deformation.

The chemical compositions of the films were determined by electron spectroscopy for chemical analysis (ESCA, PHI 5000 VersaProbe). Monochromatic Al Kα radiation was used, and X-ray beam was focused at a diameter of 100 µm and scanned on a 250 µm × 250 µm surface at an operating power of 25 W (15 kV). Photoelectron survey spectra were obtained using a hemispherical analyzer at a pass energy of 58.7 eV with an energy step of 0.4 eV. The detection limits are quoted as 0.01 to 0.5 at.% for practical analyses. Prior to measuring, the film surface was cleaned by sputtering with an Ar^+ ion beam set to a voltage of 2 kV. Crystal structures were analyzed using X-ray Diffractometer (XRD, MAC SCIENCE MPXIII, BRUKER). The system was operated at 40 kV and 30 mA with Cu K$_\alpha$ radiation. The scanning speed was 0.25°/min with the 2θ angle range of 20–80°. The rotation speed of the sample was 21,600°/min. Furthermore, from the full width at half-maximum, the average grain sizes of the coatings were calculated using Scherrer's formula:[19)] $D = K\lambda/\beta\cos\theta$, where K is the Scherrer constant, λ is the X-ray wavelength ($\lambda_{\mathrm{Cu\,K}\alpha} = 1.54$ Å), β is the full width at half-maximum, and θ is the Bragg diffraction angle. A field emission scanning electron microscope (JEOL JSM-6700F) was used to analyze film morphology and thickness. The thicknesses of the ZnMgO:Al coatings grown at RT, 200, 300, and 400 °C were 270, 317, 330, and 230 nm, respectively. A UV/visible spectrophotometer (HITACHI U-3010) was used to measure light transmittance. The electrical properties of the films were determined by performing Hall effect measurements (ECOPIA HMS-2000) under an applied magnetic field of 0.55 T. The gold coated and spring-loaded contact probes were directly contacted on the four edges of samples with a dimension of 1 cm × 1 cm.

3. Results and discussion

Table 2 presents the compositions of ZnMgO:Al coatings deposited at various substrate temperatures. As the substrate temperature increased from RT to 400 °C, the Zn ratio decreased from 83.33 to 81.65 %, whereas the Mg and Al ratios increased from 7.17 to 8.19 % and from 1.39 to 1.50 %, respectively. The slightly lower Zn ratio in the

Table 2. Elemental percentages of ZnMgO:Al coatings grown on glass as a function of substrate temperature, obtained by ESCA

Substrate temperature (°C)	Mg (at.%)	Zn (at.%)	Al (at.%)	O (at.%)	[Mg]/([Mg] + [Zn] + [Al]) × 100 %	[Zn]/([Mg] + [Zn] + [Al]) × 100 %	[Al]/([Mg] + [Zn] + [Al]) × 100 %
RT	7.17	42.82	1.39	48.61	13.95	83.33	2.71
200	7.25	42.93	1.41	48.42	14.05	83.22	2.73
300	7.57	42.85	1.43	48.16	14.60	82.65	2.76
400	8.19	43.13	1.50	47.18	15.50	81.65	2.85
Reference	8.73	45.33	0	45.95	16.15	83.85	0

Reference: undoped ZnMgO deposited at 400 °C.

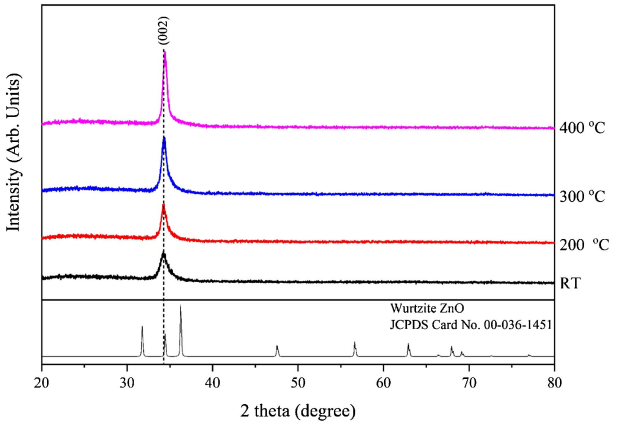


Fig. 1. XRD pattern of ZnMgO:Al coatings deposited at various substrate temperatures.

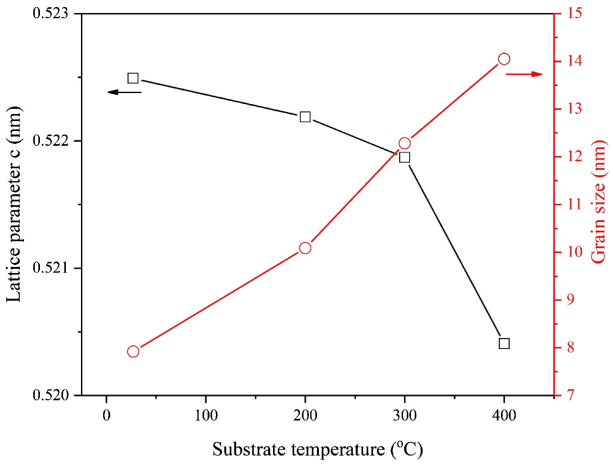


Fig. 2. Lattice parameters c and grain sizes of ZnMgO:Al coatings deposited at various substrate temperatures.

film deposited at the elevated substrate temperature was caused by Zn desorption during deposition due to the high vapor pressure of Zn.[5] Moreover, the high substrate temperature resulted in lower O_2 content, which may be attributed to the increased oxygen desorption from grain boundaries.[6),20] It is noted that the oxygen content of the undoped ZnMgO coating is lower than that of the Al-doped ZnMgO coating. This is because the nuclear charge of Al^{3+} can capture more oxygen than Zn^{2+}, leading to increase of the oxygen content in coating.[21),22]

Figure 1 shows the XRD patterns of the ZnMgO:Al coatings deposited at various substrate temperatures. The ZnMgO:Al coatings exhibited a single hexagonal wurtzite structure without significant phase separation, indicating that Mg^{2+} and Al^{3+} ions were substituted into Zn^{2+} sites or incorporated into interstitial sites in the lattice without altering the hexagonal wurtzite structure of ZnO. In terms of the preferred orientation, the ZnMgO:Al coatings possessed a strong (002) hexagonal orientation, which indicated the preferred c-axis orientation perpendicular to the substrate as the (002) plane is the lowest energy surface for the thin film.[23] Moreover, the preferred orientation strongly depends on the number of nearest neighbors offered by the growing plane to the adparticles. In the case of the hexagonal wurtzite structure, when the state of the sputtered adparticles is ZnO ad-molecule, the (002) plane offers the highest number of nearest neighbors to the incoming adatom. Thus, the ZnMgO:Al coatings grew along the c-axis orientation.[24] As the substrate temperature increased, the intensity of the (002) orientation increased due to the enhanced texture. The elevated substrate temperature leads to a stronger (002) orientation to minimize the surface energy of the film.

Figure 2 shows the lattice parameters c, stress, and grain size of the ZnMgO:Al coatings obtained from XRD patterns, which were strongly affected by the increase in the substrate temperature. The coating deposited at RT had a grain size of 7.9 nm. As the substrate temperature increased to 400 °C, the adatoms diffused more readily on the substrate surface, significantly increasing the grain size to 14.0 nm. The coating deposited at RT revealed a lattice parameter of 0.5225 nm. As the substrate temperature increased to 400 °C, the lattice parameter decreased to 0.5204 nm. The lattice shrinkage can be attributed to three reasons: (1) the substitution of Mg^{2+} (0.072 nm) and Al^{3+} ions (0.053 nm) with smaller ionic radii than that of Zn^{2+} (0.074 nm) shortens the c-axis length. Hence, interstitial defects diffuse out or are incorporated into the lattice sites at high substrate temperatures.[7] (2) The reduction in the Zn ratio is represented by the decrease in the lattice parameter. The lower Zn ratio at higher substrate temperatures means that more Mg^{2+} and Al^{3+} ions are incorporated into the coatings, thereby shortening the c-axis length. (3) In magnetron sputtering, the impact of incident energetic particles produced recoil implantation of the coating material surface atoms and entrapment of working gas atoms, which leads to the generation of compressive stress. As the substrate temperature increases, the stress release and

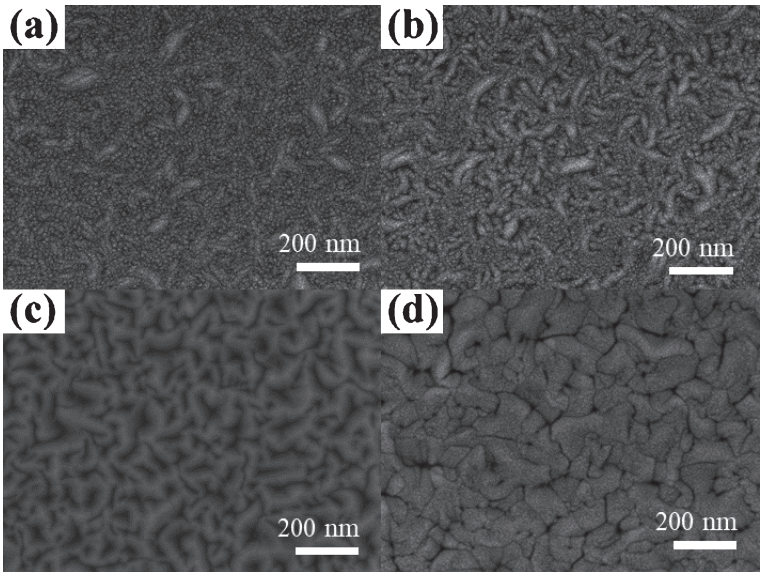


Fig. 3. SEM micrographs of ZnMgO:Al coatings deposited at various substrate temperatures: (a) RT, (b) 200 °C, (c) 300 °C, and (d) 400 °C.

Table 3. Summary of Hall effect measurement results and FOM for ZnMgO:Al coatings at various substrate temperatures. μ is the Hall mobility, N is the carrier density (electron), and ρ is the electrical resistivity

Substrate temperature (°C)	μ ($cm^2/V\,s$)	N ($\times 10^{20}\,cm^{-3}$)	ρ ($\Omega\,cm$)	FOM (Ω)
RT	0.86	0.31	1.15×10^{-1}	9.98×10^{-5}
200	1.93	0.68	2.33×10^{-2}	5.91×10^{-4}
300	3.54	1.09	8.02×10^{-3}	2.03×10^{-3}
400	4.66	2.35	2.82×10^{-3}	3.91×10^{-3}
Reference	N/A	N/A	N/A	N/A

Reference: undoped ZnMgO deposited at 400 °C.

lattice shrinkage are expected since atoms at a non-equilibrium site can escape from the surface or migrate to an equilibrium site.[25),26)]

Figure 3 shows the SEM images of ZnMgO:Al coatings deposited at various substrate temperatures. The coating deposited at RT was composed of closely packed nanocrystallites. When the substrate temperature increased to 200 °C, larger grain size and higher surface undulation were observed. With the substrate temperature increasing to 300 °C, an aggregate structure was formed due to grain coalescence, and the coating consisted of a dense array of grains. With the further increase in the substrate temperature (400 °C), the grains continued to grow. In brief, as the substrate temperature increased, the grain size increased, consistent with the XRD results.

The Hall mobility (μ), carrier concentration (N), electrical resistivity (ρ), and figure of merit (FOM) as a function of substrate temperature are shown in **Table 3**. The FOM is defined as $\Phi = T^{10}/R_{sh}$ in this study, where T is the average transmittance in the visible region, and R_{sh} is the sheet resistance.[27)] For the undoped ZnMgO coating, no measurable resistivity was detected within the measurement limit of the Hall effect instrument. All the coatings were degenerately doped n-type semiconductors. As the substrate temperature increased from RT to 400 °C, the resistivity decreased from 1.15×10^{-1} to 2.82×10^{-3} $\Omega\,cm$, whereas the carrier concentration increased from 0.31×10^{20} to $2.35 \times 10^{20}\,cm^{-3}$. The mobility also increased from 0.86 to $4.66\,cm^2/V\,s$. The polycrystalline TCO coatings were affected mainly by the carrier concentration and mobility. The low mobility of the ZnMgO:Al coating deposited at a low substrate temperature is caused by the obstruction of carrier movement due to poor crystallinity and small grain size.[28)] The crystallinity of the coating increased with the increased substrate temperature, leading to higher mobility. The carrier concentration demonstrated the same trend as the Hall mobility. As mentioned previously, the high substrate temperature can promote oxygen desorption from grain boundaries and help more Al^{3+} ions to diffuse into the lattice sites, thereby increasing the carrier concentration.[7)] These results suggest that Al^{3+} ions are stably substituted in the ZnO coating as sufficient thermal energy promoted their diffusion. In addition, the increased carrier concentration of annealed coatings with increased substrate temperature can be associated with chemical composition. As specified in the abovementioned ESCA analysis, with the increase in the substrate temperature, more oxygen vacancies were formed, which could serve as donors. Moreover, the high substrate temperature also increased the Al content in the coatings. Accordingly, more Al substitution occurred, which could donate two electrons. These phenomena greatly increased the carrier concentration of the coatings.

Figure 4(a) shows the visible transmittance spectra of

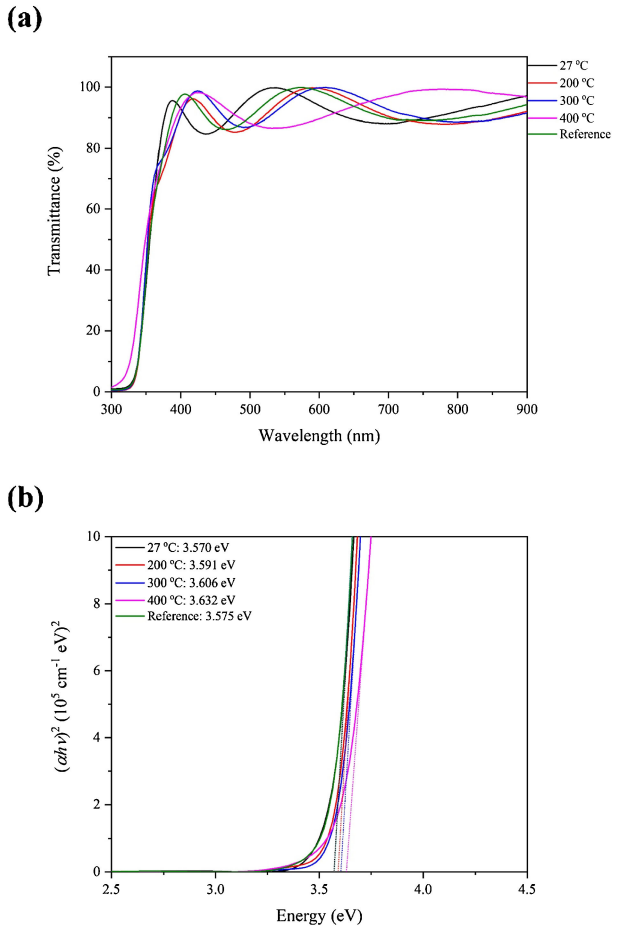


Fig. 4. (a) Transmittance spectra of ZnMgO:Al coatings deposited at various substrate temperatures and (b) variation in $(\alpha h\nu)^2$ with $h\nu$.

ZnMgO:Al coatings deposited at various substrate temperatures. The average optical transmittance in the visible range (380–780 nm) exceeded 91 % for all coatings, and a sharp fundamental absorption edge was found in the UV region. Hence, the substrate temperature did not significantly affect the transparency of the coatings in the visible range. However, as the substrate temperature increased, the absorption edge shifted to smaller wavelengths. The optical bandgap (E_g) of the films can be determined by plotting $(\alpha h\nu)^2$ versus $h\nu$ and extrapolating the straight-line portion of the energy axis [Fig. 4(b)], where $h\nu$ is the incident photon energy, and α is the absorption coefficient.[29] Figure 4(b) shows that the optical bandgaps of the ZnMgO:Al coatings deposited at RT, 200, 300, and 400 °C were 3.570, 3.591, 3.606, and 3.632 eV, respectively. The optical bandgap increased with the substrate temperature. The larger optical bandgap of the coatings deposited at elevated substrate temperatures is due to the Burstein–Moss effect; that is, an increase in the Fermi level in the conduction band of a semiconductor leads to an energy band-widening effect.[30),31] The widening of the bandgap is related to the increase in carrier concentration and corresponds to low electrical resistivity. In addition, the increase in bandgap may be attributed to chemical composition variations at different substrate temperatures. The Zn desorption at a high substrate temperature causes a corresponding increase in Mg content, which is responsible for the increased bandgap. Noticeably, the undoped ZnMgO coating heated at 400 °C showed a smaller bandgap (3.575 eV) than that with 2.0 wt.% Al_2O_3 because of the Burstein–Moss effect. Therefore, the results indicate that ZnMgO:Al optoelectronic devices can operate well in the UV region.

In this study, ZnMgO:Al coatings were deposited via sputtering using single ceramic targets. The high substrate temperature promoted grain growth and significantly improved the effectiveness of Al doping. Based on the FOMs of the coatings deposited at various substrate temperatures, the coating deposited at 400 °C was optimal, exhibiting an electrical resistivity of 2.82×10^{-3} Ω cm and optical transmittance of 91 %, suitable for large-area UV optoelectronic device applications, which is comparable with available wide-bandgap TCOs recently published, as shown in Table 1. Our study shows that enhancing the electrical performance of ZnMgO coatings can be enabled by Al doping and substrate heating during deposition.

4. Conclusions

In this study, we investigated ZnMgO:Al coatings deposited by RF magnetron sputtering at substrate temperatures from RT to 400 °C. XRD analysis confirmed that the coatings exhibited the same hexagonal wurtzite structure with a strong (002) *c*-axis orientation. Increasing the substrate temperature promoted adatom diffusion and increased the grain size, which reduced grain boundary scattering and increased the Hall mobility. The high substrate temperature improved the effectiveness of Al doping. The desorption behavior of Zn and O at high substrate temperatures increased the concentrations of Al. Hence, the ZnMgO:Al coating deposited at 400 °C showed the lowest electrical resistivity (2.82×10^{-3} Ω cm) because of the combined effects of the increased Hall mobility and carrier concentration. The optical bandgap varied from 3.570 to 3.632 eV with the increased substrate temperature, which was attributed to the Burstein–Moss effect and increased Mg content in films. The wide bandgap, high transparency, and moderate electrical conductivity of ZnMgO:Al coatings make them ideal transparent electrodes in UV photovoltaic devices.

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